**Department of Computer Science and Engineering**
**Academic Year 2021– 2022(Odd Semester)**
**Degree, Semester & Branch: III Semester B.E. ECE-'B' Section**
**Course Code & Title: EC8393 Fundamentals of Data Structures in C**
**Name of the Faculty member (s): Dr.V. Anusuya, ASCP/CSE**

## Innovative Practice Description

- **Unit / Topic: Unit 3 /** Evaluation of Expression
- **Course Outcome:** The student will be able to apply appropriate linear data structure operations for any application using C
- **Topic Learning Outcome:** TLO9.
- **Activity Chosen: Think Pair Share**
- **Justification:**
    - After explaining the concept of evaluation of expression such as infix to postfix conversion, infix to prefix and postfix evaluation in stack application, to understand the learning level of the student about evaluating the expression this activity was given.
- **Time Allotted for the Activity:** 15 Minutes
- **Details of the Implementation:**
    - Displayed the different questions in the topics infix to postfix conversion, infix to prefix and postfix evaluation for each pair of students in the class as shown in Figure 1.
    - Identified 25 pair of students and asked them to think for five minute about evaluation of expression.
    - After that, the students were asked to discuss the answer for the questions in the Dias as shown in the figure 2,3,4,5.

**CO – PO / PSO mapping:**

| CO | PO1 | PO2 | PO3 | PO4 | PSO1 |
|---|---|---|---|---|---|
| **CO2** | 3 | 3 | 2 | 2 | 2 |

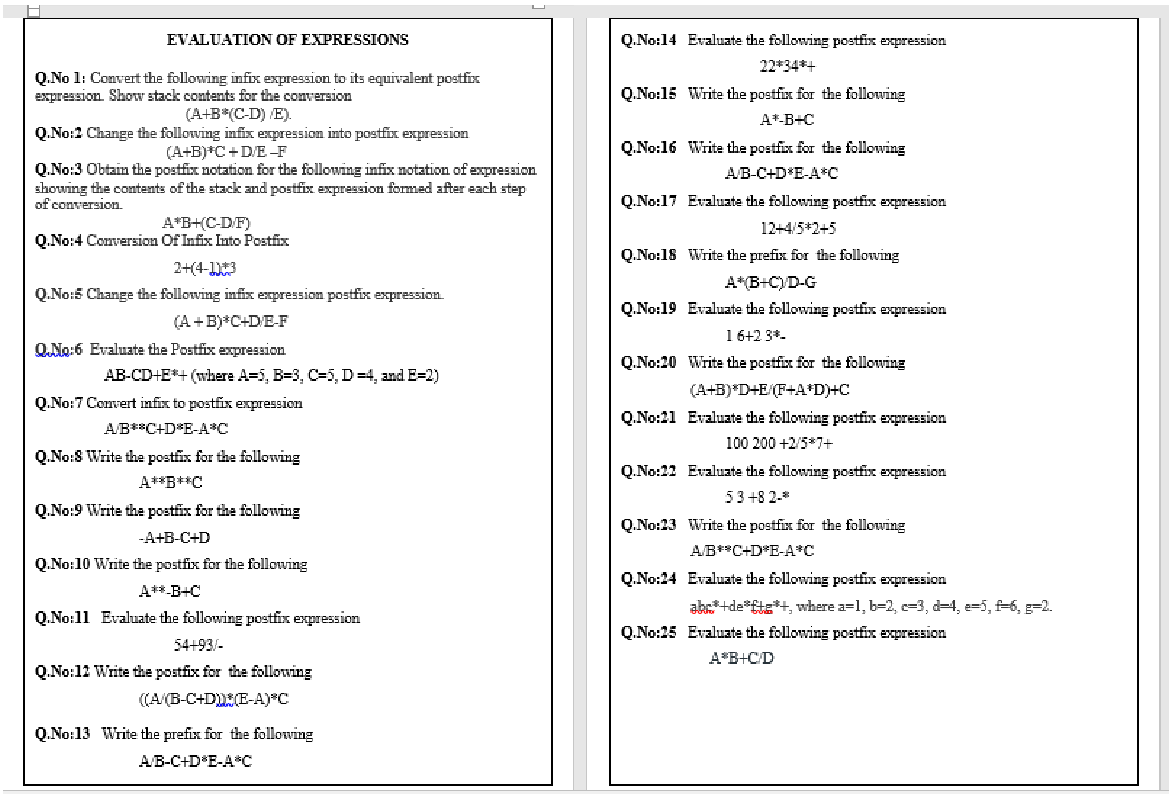
**(1 – Low 2 – Moderate 3 – High)**

- **PO / PSO mapped:**

| **Innovative practice** | **PO1** | **PO2** | **PO3** | **PO4** | **PSO1** |
|---|---|---|---|---|---|
| | 3 | 3 | 2 | 2 | 2 |
| **Justification for correlation** | The basic knowledge of engineering fundamentals is necessary to | The student should identify analyze problem to evaluate expression | The student will be able to design and find the | | The student will be able to apply stack application in the field of information and Communication Technology |

| | | | | | |
|---|---|---|---|---|---|
| | understand the concept of expression evaluation | | solution for the problem | | |

- **Images / Screenshot of the practice:**



EVALUATION OF EXPRESSIONS

**Q.No 1:** Convert the following infix expression to its equivalent postfix expression. Show stack contents for the conversion
(A+B*(C-D) /E).

**Q.No:2** Change the following infix expression into postfix expression
(A+B)*C + D/E –F

**Q.No:3** Obtain the postfix notation for the following infix notation of expression showing the contents of the stack and postfix expression formed after each step of conversion.
A*B+(C-D/F)

**Q.No:4** Conversion Of Infix Into Postfix
2+(4-1)*3

**Q.No:5** Change the following infix expression postfix expression.
(A + B)*C+D/E-F

**Q.No:6** Evaluate the Postfix expression
AB-CD+E*+ (where A=5, B=3, C=5, D =4, and E=2)

**Q.No:7** Convert infix to postfix expression
A/B**C+D*E-A*C

**Q.No:8** Write the postfix for the following
A**B**C

**Q.No:9** Write the postfix for the following
-A+B-C+D

**Q.No:10** Write the postfix for the following
A**-B+C

**Q.No:11** Evaluate the following postfix expression
54+93/-

**Q.No:12** Write the postfix for the following
((A/(B-C+D))*(E-A)*C

**Q.No:13** Write the prefix for the following
A/B-C+D*E-A*C

**Q.No:14** Evaluate the following postfix expression
22*34*+

**Q.No:15** Write the postfix for the following
A*-B+C

**Q.No:16** Write the postfix for the following
A/B-C+D*E-A*C

**Q.No:17** Evaluate the following postfix expression
12+4/5*2+5

**Q.No:18** Write the prefix for the following
A*(B+C)/D-G

**Q.No:19** Evaluate the following postfix expression
1 6+2 3*-

**Q.No:20** Write the postfix for the following
(A+B)*D+E/(F+A*D)+C

**Q.No:21** Evaluate the following postfix expression
100 200 +2/5*7+

**Q.No:22** Evaluate the following postfix expression
5 3 +8 2-*

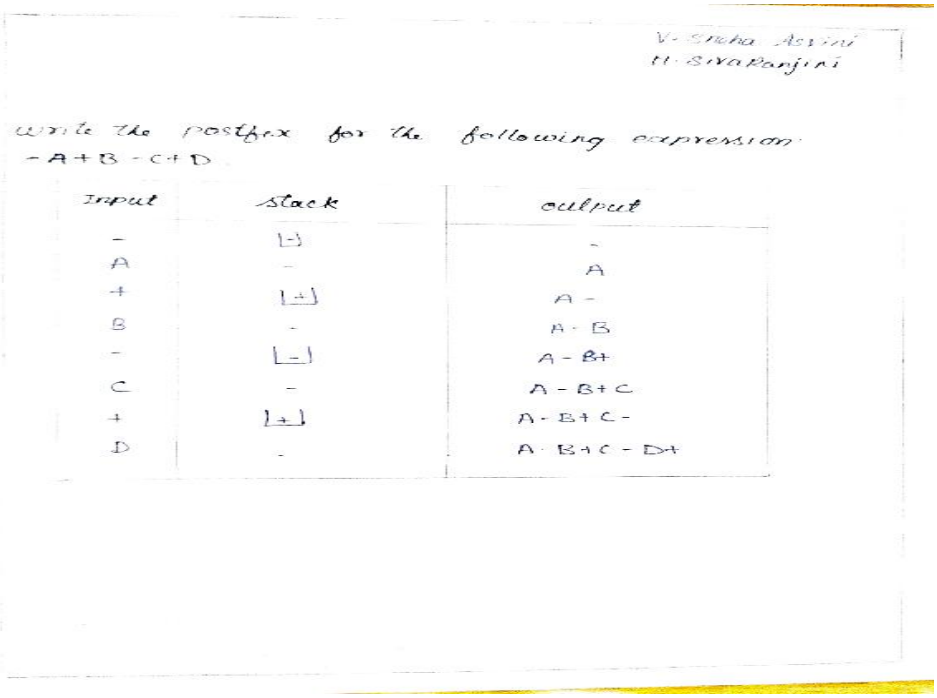
**Q.No:23** Write the postfix for the following
A/B**C+D*E-A*C

**Q.No:24** Evaluate the following postfix expression
abc*+de*f+g*+, where a=1, b=2, c=3, d=4, e=5, f=6, g=2.

**Q.No:25** Evaluate the following postfix expression
A*B+C/D

**Figure 1 Questions given to students**

**Screen shots**



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write the postfix for the following expression
- A + B - C + D

| Input | Stack | output |
|---|---|---|
| - | [-] | - |
| A | - | A |
| + | [+] | A - |
| B | - | A - B |
| - | [-] | A - B+ |
| C | - | A - B + C |
| + | [+] | A - B + C - |
| D | - | A - B + C - D+ |

**Figure 2. Sample Answers of students**

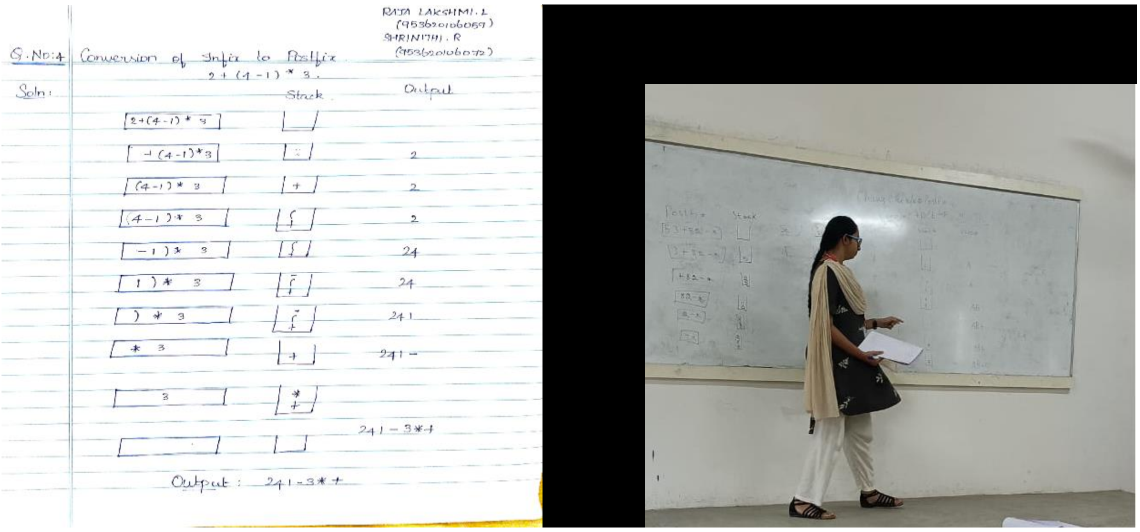


**Figure 3. Sample Answers of students**

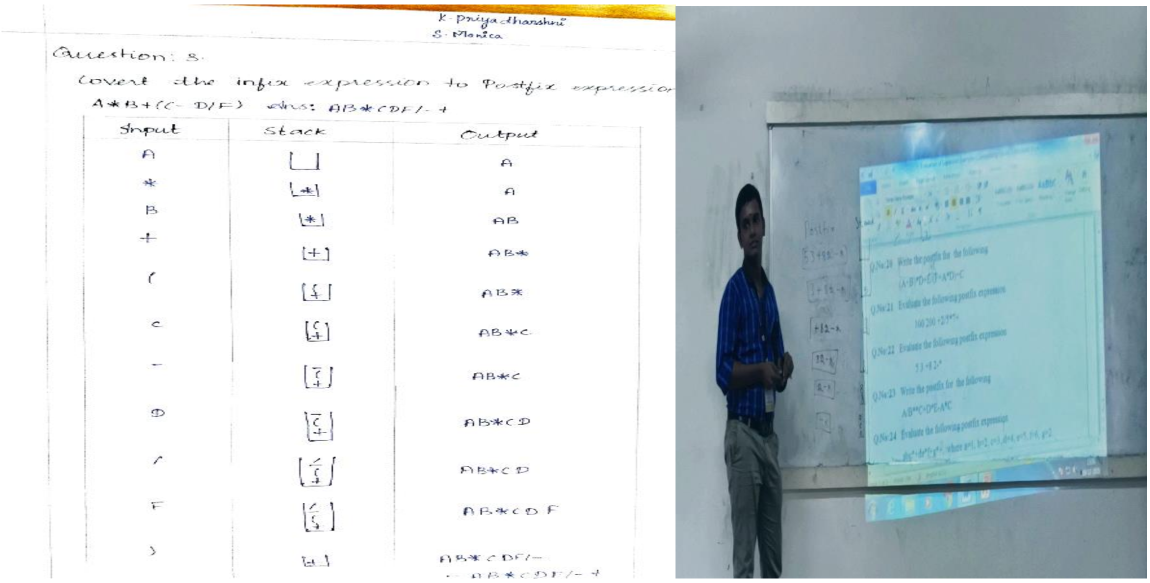


**Figure 4. Sample Answers of students**

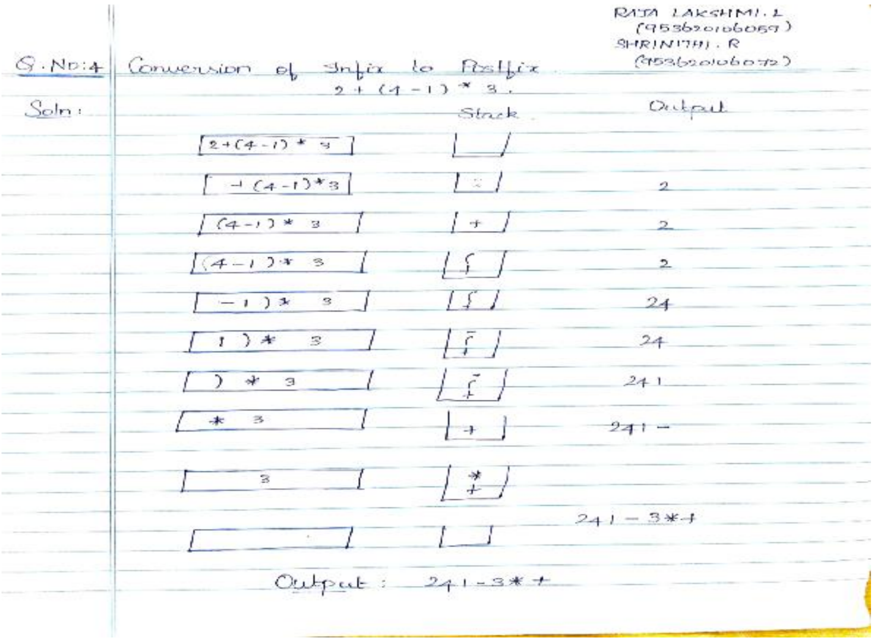


**Figure 5. Sample Answers of students**

- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    - The students are actively participated in the activity.
    - The student said this activity helped to think and recall information and improved our problem solving skills.
  - ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
    - Most of the students gave the right answer for the question allotted to them
    - Interaction with the faculty member and their class mates during the class was effective.
    - The fear of students was reduced compare when discussing with their friends.
    - It helps the students to think individually about a topic or answer to a given question.
    - From this activity, the students can get more clarity in the expression evaluation by discussing and sharing their views with the other students in the class.

❖ ***Challenges faced in implementation:***

Some students are hesitated and lack to share the concepts that they learnt in the class.

References:


- https://www.ritrjpm.ac.in/images/computer-science/


**Signature of Faculty Member** **HOD**